

SEMESTER-WISE DISTRIBUTION OF COURSES

I Semester			
S. No	Course Code	Course Title	Credit Hours
1.	DIKSHA-111	Deeksharambh (Induction cum Foundation Course)	1 Week Non-Gradial
2.	LPM-111	Poultry Production Technology*	2(0+2)
3.	PPATH-111	Mushroom Production Technology*	2(0+2)
4.	AGEXT-111	Rural Sociology & Educational Psychology	2(2+0)
5.	AGEXT-112	Communication Skills	2(1+1)
6.	AGRON-111	Fundamentals of Agronomy	3(2+1)
7.	AGRON-112	Farming Based Livelihood System	3(2+1)
8.	HORT-111	Fundamentals of Horticulture	3(2+1)
9.	SSAC-111	Fundamentals of Soil Science	3(2+1)
10.	NCC I/ NSS I/ PHEDI	NSS/NCC/Physical Education & Yoga Practices	1(0+1)
11.	MATH/BIO-111	Introductory Mathematics/ Introductory Biology**	1(1+0)
Total Credit			21(11+10)+1**
*Skill Enhancement Courses; **NC: Non-gradial courses			

Induction cum Foundation Course (Deekshaarambh)-Non-gradiel
DIKSHA-111 **Credit Hours: 1(1+0)**

Objectives:

- Help for cultural Integration of students from different backgrounds,
- Know about the operational framework of academic process in the University/College/Institute
- Instilling life and social skills,
- Social Awareness, Ethics and Values, Team Work, Leadership, Creativity, etc.
- Identify the traditional values and indigenous cultures along with diverse potentialities both in indigenous and developed scenario.
- Identify strength and weakness of the students in different core areas of the discipline.

The details of activities will be decided by the parent universities. The structure shall include, but not restricted to:

- i. Discussions on operational framework of academic process in the University, as well as interactions with academic & research managers of the University
- ii. Interaction with alumni, business leaders, perspective employers, outstanding achievers in related fields, and people with inspiring life experiences
- iii. Group activities to identify the strength and weakness of students (with expert advice for their improvement) as well as to create a platform for students to learn from each other's life experiences
- iv. Activities to enhance cultural integration of students from different backgrounds.
- v. Field visit to related fields/ establishments
- vi. Sessions on personality development (instilling life and social skills, social awareness, ethics and values, team work, leadership, etc.) and communication skills.

LPM-111 POULTRY PRODUCTION TECHNOLOGY 2 (0+2)

Objective:

1. To make the students aware about poultry based livelihood system in agriculture.
2. To expose the student to various aspects of establishment and management of a poultry farm.
3. To enhance the skill for employment and entrepreneurship in poultry production.

Practical

External body parts of poultry birds. Identification of different breeds of poultry. Handling and Identification methods of poultry. Visit to poultry farms. Daily routine farm operations and farm

records of poultry farms. Judging, selection & culling of poultry. Planning and layout of housing for different types of poultry. Computation and formulation of ration for poultry. Study composition and nutritive composition of poultry eggs. Collection, Grading, Handling and storage of eggs. Hatchery management of poultry. Management of chicks, growers and layers. Poultry farm equipment's. Debeaking, dusting of poultry. Different diseases of livestock and poultry, their prevention and control. Economics of poultry production.

Lecture Schedule: Practical

S. No.	Topic	No. of Lecture
1.	External body parts of poultry birds.	1
2.	Identification of different breeds of poultry	2
3.	Handling and Identification methods of poultry.	2
4.	Visit to poultry farms	1
5.	Daily routine farm operations and farm records of poultry farms.	2
6.	Judging, selection & culling of poultry.	2
7.	Planning and layout of housing for different types of poultry.	3
8.	Computation and formulation of ration for poultry.	2
9.	Composition of poultry eggs	2
10.	Collection, Grading, Handling and storage of eggs	2
11.	Hatchery management	2
12.	Management of chicks, growers and layers.	2
13.	Poultry farm equipment	2
14.	Debeaking, dusting and culling of poultry	2
15.	Different diseases of livestock and poultry, their prevention and control.	3
16.	Economics of broiler and layer production.	2
	Total practical classes	32

Suggested Readings:

1. A Textbook of Animal Husbandry by G. C Banerjee (8th Edition)
2. Livestock Production Management by N.S.R. Sastry and C.K. Thomas (2021)
3. Handbook of Animal Husbandry by ICAR (Latest)
4. Textbook of Commercial Poultry production and Hatchery Management by Dr. M.

PPATH- 111**MUSHROOM PRODUCTION TECHNOLOGY****2(0+2)****Objectives:**

1. Provide a comprehensive understanding of the biology, life cycle, and classification of mushrooms, including edible, medicinal, and industrially significant species.
2. Teach advanced methods for mushroom cultivation, including substrate preparation, spawning, maintenance of environmental conditions, pest and disease management, and post-harvest handling.
3. Encourage eco-friendly and sustainable approaches to mushroom production by utilizing agricultural waste and low-cost production techniques.
4. Foster entrepreneurship by guiding learners on business planning, marketing, and the economic potential of mushroom farming.
5. Train participants to identify and resolve common challenges in mushroom production, ensuring consistent quality and yield.

Practical:

Current status and scope of mushroom cultivation in India. Important features of edible fungi. Nutritional and medicinal value of mushrooms. Preparation of media and culture preparation. Sub-culturing for culture maintenance and its preservation. Spawn preparation techniques. Collection of wild mushroom flora of India. Raw material formulations for *Agaricus bisporus* (button mushroom). Composting (long method and short method). Casing preparation and crop management practices. Mushroom farm design and infrastructure required for commercial unit. Cultivation techniques of *Pleurotus florida* (dhingri), *Lentinus edodes* (shiitake), *Calocybe indica* (milky) and *Volvoriellavolvacea* (paddy straw). Marketing of mushrooms. Mushroom diseases and their control. Economics of mushrooms and exposure visit to commercial farms.

Lecture Schedule: Practical

S. No.	Topic	No of Lectures
1.	Current status and scope of mushroom cultivation in India	1
2.	Important features of edible fungi	1

3.	Nutritional and medicinal value of mushrooms	1
4.	Preparation of media	1
5.	Culture preparation	1
6.	Sub-culturing for culture maintenance and its preservation	1
7.	Spawn preparation techniques	2
8.	Collection of wild mushroom flora of India	1
9.	Raw material formulations for <i>Agaricus bisporus</i> (button mushroom)	2
10.	Composting (long method and short method)	2
11.	Casing preparation	1
12.	Crop management practices	1
13.	Mushroom farm design and infrastructure required for commercial unit	2
14.	Cultivation techniques of <i>Pleurotus florida</i> (dhingri)	2
15.	Cultivation techniques of <i>Lentunus edodes</i> (shiitake)	2
16.	Cultivation techniques of <i>Calocybe indica</i> (milky)	2
17.	Cultivation techniques of <i>Volvariellavolvacea</i> (paddy straw)	2
18.	Marketing of mushrooms	1
19.	Mushroom diseases and their control	2
20.	Preparation of value-added products from mushrooms	2
21.	Economics of mushrooms	1
22.	Exposure visit to commercial farms	1
Total		32

AGEXT- 111 RURAL SOCIOLOGY AND EDUCATIONAL PSYCHOLOGY 2 (2+0)

Objective

Provide knowledge on concept and importance of sociology and rural sociology as well as the relationship with Extension Education

Theory

Extension Education and Agricultural Extension: Meaning, definition, scope, and importance. Sociology and rural sociology: Meaning, definition, scope, importance of rural sociology in Agricultural Extension, and interrelationship between rural sociology and Agricultural Extension. Indian Rural Society:

important characteristics, differences and relationship between rural and urban societies. Social Groups: Meaning, definition, classification, factors considered information and organization of groups, motivation in group formation and role of social groups in Agricultural Extension. Social Stratification: Meaning, definition, functions, basis for stratification, forms of social stratification- characteristics and differences between class and caste system. Cultural concepts: culture, customs, folkways, mores, taboos, rituals. Traditions: Meaning, definition and their role in Agricultural Extension. Social Values and Attitudes: Meaning, definition, types and role of social values and attitudes in agricultural Extension. Social Institutions: Meaning, definition, major institutions in rural society, functions, and their role in agricultural Extension. Social Organizations: Meaning, definition, types of organizations and role of social organizations in agricultural Extension. Social Control: Meaning, definition, need of social control and means of social control. Social change: Meaning, definition, nature of social change, dimensions of social change and factors of social change. Leadership: Meaning, definition, classification, roles of leader, different methods of selection of professional and lay leaders. Training of Leaders: Meaning, definition, methods of training, Advantages and limitations in use of local leaders in Agricultural Extension, Psychology and educational psychology: Meaning, definition, scope, and importance of educational psychology in Agricultural Extension. Intelligence: Meaning, definition, types, factors affecting intelligence and importance of intelligence in Agricultural Extension. Personality: Meaning, definition, types, factors influencing the personality and role of personality in agricultural Extension. Teaching: Learning process: Meaning and definition of teaching, learning, learning experience and learning situation, elements of learning situation and its characteristics. Principles of learning and their implication of teaching.

Lecture Schedule:Theory

S.N.	Topic	No. of Lectures
1.	Extension Education and Agricultural Extension: Meaning, Definition, Scope, and Importance	1
2.	Sociology and Rural Sociology: Meaning, Definition, Scope, Importance of Rural Sociology in Agricultural Extension, and Interrelationship between Rural Sociology and Agricultural Extension	2
3.	Indian Rural Society: Important Characteristics, Differences and Relationship between Rural and Urban Societies	2
4.	Social Groups: Meaning, Definition, Classification, Factors Considered in Formation and Organization of Groups	2
5.	Motivation in Group Formation and Role of Social Groups in Agricultural	1

	Extension	
6.	Social Stratification: Meaning, Definition, Functions, Basis for Stratification, Forms of Social Stratification-Characteristics and-Differences between Class and Caste System	2
7.	Cultural Concepts: Culture, Customs, Folkways, Mores, Taboos, Rituals, and Traditions: Meaning, Definition and their Role in Agricultural Extension	2
8.	Social Values and Attitudes: Meaning, Definition, Types and Role of Social Values and Attitudes in Agricultural Extension	2
9.	Social Institutions: Meaning, Definition, Major Institutions in Rural Society, Functions, and their Role in Agricultural Extension	2
10.	Social Organizations: Meaning, Definition, Types of Organizations and Role of Social Organizations in Agricultural Extension	2
11.	Social Control: Meaning, Definition, Need of Social Control and Means of Social Control	2
12.	Social Change: Meaning, Definition, Nature of Social Change, Dimensions of Social Change and Factors of Social Change	2
13.	Leadership: Meaning, Definition, Classification, Roles of Leader, Different Methods of Selection of Professional and Lay Leaders	2
14.	Training of Leaders: Meaning, Definition, Methods of Training, Advantages and Limitations in Use of Local Leaders in Agricultural Extension	1
15.	Psychology and Educational Psychology: Meaning, Definition, Scope, and Importance of Educational Psychology in Agricultural Extension	1
16.	Intelligence: Meaning, Definition, Types, Factors affecting Intelligence and Importance of Intelligence in Agricultural Extension	1
17.	Personality: Meaning, Definition, Types, Factors Influencing the Personality and Role of Personality in Agricultural Extension	2
18.	Teaching: Learning process: Meaning and Definition of Teaching, Learning, Learning Experience and Learning Situation, Elements of Learning Situation and its Characteristics	2
19.	Principles of Learning and their Implication of Teaching	1

	Total	32
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Suggested Readings:

1. R. Desai -Rural Sociology in India
2. Dahama O. P. and Bhatnagar, O. P. - Education and Communication for Development
3. J.B. Chitambar -Introductory Rural Sociology
4. M.B. Ghorpade- Essential of psychology
5. Prepared You Tube videos
6. R Velusamy Textbook on Rural Sociology and Educational Psychology
7. Ray, G. L. -Extension Communication and Management
8. Sandhu A. S. -Textbook on Agricultural Communication
9. Web Materials

AGEXT- 112**COMMUNICATION SKILLS****2 (1+1)****Objectives**

To acquire competence in oral, written and non-verbal communication, develop strong personal and professional communication and demonstrate positive group communication.

Theory

Communication Process: The magic of effective communication; Building self-esteem and overcoming fears; Concept, nature and significance of communication process; Meaning, types and models of communication; Verbal and non-verbal communication; Linguistic and non-linguistic barriers to communication and reasons behind communication gap/ miscommunication. Basic Communication Skills: Listening, Speaking, Reading and Writing Skills; Precs writing/Abstracting/Summarizing; Style of technical communication Curriculum vitae/resume writing; Innovative methods to enhance vocabulary, analogy questions.

Structural and Functional Grammar: Sentence structure, modifiers, connecting words and verbal; phrases and clauses; Case: subjective case, possessive case; objective case; Correct usage of nouns, pronouns and antecedents, adjectives, adverbs and articles; Agreement of verb with the subject: tense, mood, voice; Writing effective sentences; Basic sentence faults;

Lecture Schedule;

S. N.	Topic	No. of Lectures
1.	Communication Process: The Magic of Effective Communication; Building Self-Esteem and Overcoming Fears; Concept, Nature and Significance of Communication Process; Meaning, Types and Models of Communication	2
2.	Verbal and Non-verbal Communication; Linguistic and Non-linguistic	2

	Barriers to Communication and Reasons behind Communication Gap/ Miscommunication	
3.	Communication Skills: Listening, Speaking, Reading and Writing Skills	2
4.	Precis Writing/Abstracting/Summarizing	1
5.	Style of Technical Communication: Curriculum Vitae/Resume Writing	2
6.	Innovative Methods to Enhance Vocabulary, Analogy Questions	1
7.	Structural and Functional Grammar: Sentence Structure, Modifiers, Connecting Words and Verbal; Phrases and Clauses	2
8.	Case: Subjective Case, Possessive Case; Objective Case	1
9.	Correct Usage of Nouns, Pronouns and Antecedents, Adjectives, Adverbs and Articles	1
10.	Agreement of Verb with the Subject: Tense, Mood, Voice	1
11.	Writing Effective Sentences; Basic Sentence Faults	1
Total		16

Practical

Listening and note taking; Writing skills: precis writing, summarizing and abstracting; Reading and comprehension (written and oral) of general and technical articles; Micro-presentations and Impromptu Presentations: Feedback on presentations; Stage manners: grooming, body language, voice modulation, speed; Group discussions; Public speaking exercises; vocabulary building exercises; Interview Techniques; organization of events.

Lecture Schedule;

S. N.	Topic	No. of Lectures
1.	Communication Process: The Magic of Effective Communication; Building Self-Esteem and Overcoming Fears; Concept, Nature and Significance of Communication Process; Meaning, Types and Models of Communication	2
2.	Verbal and Non-verbal Communication; Linguistic and Non-linguistic Barriers to Communication and Reasons behind Communication Gap/ Miscommunication	2
3.	Communication Skills: Listening, Speaking, Reading and Writing	2

	Skills	
4.	Precis Writing/Abstracting/Summarizing	1
5.	Style of Technical Communication: Curriculum Vitae/Resume Writing	2
6.	Innovative Methods to Enhance Vocabulary, Analogy Questions	1
7.	Structural and Functional Grammar: Sentence Structure, Modifiers, Connecting Words and Verbal; Phrases and Clauses	2
8.	Case: Subjective Case, Possessive Case; Objective Case	1
9.	Correct Usage of Nouns, Pronouns and Antecedents, Adjectives, Adverbs and Articles	1
10.	Agreement of Verb with the Subject: Tense, Mood, Voice	1
11.	Writing Effective Sentences; Basic Sentence Faults	1
Total		16

Practical

S. N.	Topic	No. of Lectures
1.	Listening and Note Taking	2
2.	Writing Skills: Precis Writing, Summarizing and Abstracting	2
3.	Reading and Comprehension (Written and Oral) of General and Technical Articles	2
4.	Micro-presentations and Impromptu Presentations	2
5.	Feedback on Presentations	1
6.	Stage Manners: Grooming, Body Language, Voice Modulation, Speed	1
7.	Group Discussions	2
8.	Public Speaking Exercises	1
9.	Vocabulary Building Exercises	1
10.	Organization of Events	2
Total		16

Suggested readings

1. Allport, G. W. 1937. Personality: A Psychological Interpretation. Holt, New York.
2. Brown Michele and Gyles Brandreth. 1994. How to Interview and be Interviewed. Sheldon Press, London.
3. Carnegie Dale. 1997. The Quick and Easy Way to Effective Speaking. Pocket Books, New York.
4. Francis Peter S J. 2012. Soft Skills and Professional Communication. Tata McGraw Hill, New Delhi.
5. Kumar S and Pushpa Lata. 2011. Communication Skills. Oxford University Press.
6. Neuliep James W. 2003. Intercultural Communication A Contextual Approach. Houghton Mifflin Co Boston.
7. Pease, Allan. 1998. Body Language. Sudha Publications, Delhi.

8. Raman M and Singh P. 2000. Business Communication. Oxford University Press.
9. Seely J. 2013. Oxford Guide to Effective Writing and Speaking. Oxford University Press.
10. Thomson A J and Martinet A V. 1977. A Practical English Grammar. Oxford University

AGRON-111 FUNDAMENTALS OF AGRONOMY

3(2+1)

Objectives

To impart the basic and fundamental knowledge of Agronomy

Theory

Agronomy and its scope: Definition, meaning and scope of Agronomy; art, science and business of crop production, relation of Agronomy with other disciplines of Agricultural Science, fields crops and classification, importance, ecology and ecosystem. Dormancy; its type and methods of breaking dormancy. Seeds and sowing: Definitions of crops, variety and seed. Factors affecting crop stands establishment: good quality seed, proper tillage, time of sowing seed rate, depth and method of sowing: broadcasting, drilling, dibbling, transplanting etc. Tillage and tilth: Definition, objectives, types, advantages and disadvantages of tillage including conservation tillage. Crop density and geometry: plant geometry and planting geometry, its effect on growth, yield.

Crop nutrition: Definition of essential nutrients, criteria of essentiality, functional elements, classification of essential nutrients, role of macro and micro nutrients. Nutrient absorption, active and passive absorption of nutrients, forms of plant nutrients absorbed by plants, Combined /uncombined forms. Manures and fertilizers, nutrient use efficiency: Sources of nutrients: Inorganic (fertilizers), organic (manures) and bio-fertilizers; their classification and characteristics, method of preparation and role of organic manures in crop production. Integrated Nutrient Management (INM): Meaning, different approaches and advantages of INM. Green manure- role in crop production: Definition, objectives types of green manuring, desirable characteristics, advantages and limitations of green manuring.

Water management: Water resources of the world, India and the state; Soil Moisture constants: gravitational water, capillary water, hygroscopic water, Soil moisture constants.

Weeds: Definition, Importance and basics of classification of weeds and their control. Agroclimatic zones of India and the state, cropping systems: Factors affecting cropping systems, major cropping patterns and systems in the country. Sustainable crop production: Definition, importance and practices, natural resources and conservation pollution and pollutants, Allelopathy: Meaning and importance in crop production, Growth and development of crops: Definition, Meaning and factors affecting growth and development.

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Agronomy and its scope: Definition, meaning and scope of Agronomy; art, science and business of crop production	2
2.	Relation of Agronomy with other disciplines of Agricultural Science	1
3.	Fields crops and classification, importance, ecology and ecosystem	1
4.	Dormancy; its type and methods of breaking dormancy	1
5.	Seeds and sowing: Definitions of crops, variety and seed.	1
6.	Factors affecting crop stands establishment: good quality seed, proper tillage, time of sowing seed rate	1
7.	Depth and method of sowing: broadcasting, drilling, dibbling, transplanting etc.	1
8.	Tillage and tilth: Definition, objectives, types	1
9.	Advantages and disadvantages of tillage including conservation tillage	1
10.	Crop density and geometry: plant geometry and planting geometry, its effect on growth, yield	1
11.	Crop nutrition: Definition of essential nutrients, criteria of essentiality, functional elements	1
12.	Classification of essential nutrients, role of macro and micro nutrients	2
13.	Nutrient absorption, active and passive absorption of nutrients, forms of plant nutrients absorbed by plants, Combined /uncombined forms	2
14.	Manures and fertilizers, nutrient use efficiency: Sources of nutrients: Inorganic (fertilizers), organic (manures) and bio-fertilizers; their classification and characteristics, method of preparation and role of organic manures in crop production	2
15.	Integrated Nutrient Management (INM): Meaning, different approaches and advantages of INM	2
16.	Green manure- role in crop production: Definition, objectives types of green manuring, desirable characteristics, advantages and limitations of green manuring	2
17.	Water management: Water resources of the world, India and the state;	1
18.	Soil Moisture constants: gravitational water, capillary water, hygroscopic	1

	water, Soil moisture constants	
19.	Weeds: Definition, Importance and basics of classification of weeds and their control	2
20.	Agroclimatic zones of India and the state, cropping systems: Factors affecting cropping systems, major cropping patterns and systems in the country	2
21.	Sustainable crop production: Definition, importance and practices, natural resources and conservation pollution and pollutants	2
22.	Allelopathy: Meaning and importance in crop production	1
23.	Growth and development of crops: Definition, Meaning and factors affecting growth and development	1
Total		32

Practical

A visit to Instructional Crop farm and study on field crops, Identification of crops, seeds, fertilizers, pesticides, Crops and cropping systems in different Agro-climatic zones of the state, Study of some preparatory tillage implements, Study of inter tillage implements, Practice of ploughing / puddling, Study and practice of inter cultivation in field crops, Numerical exercises on calculation of seed, plant population and fertilizer requirement, Study of yield contributing characters and yield estimation of crops, Identification of weeds in different crops, Seed germination and viability test of seed, Practice on time and method of application of manures and fertilizers. Measurement of soil moisture by gravimetric and volumetric method and bulk density, Determination of field capacity, Determination of gross and net irrigation requirement. Determination of infiltration rate.

Lecturer schedule: Practical

S. No	Topic	No. of lectures
1	A visit to Instructional Crop farm and study on field crops	1
2	Identification of crops, seeds, fertilizers, pesticides	1
3	Crops and cropping systems in different Agro-climatic zones of the state,	1
4	Study of some preparatory tillage implements	1
5	Study of inter tillage implements	1
6	Practice of ploughing / puddling	1

7	Study and practice of inter cultivation in field crops	1
8	Numerical exercises on calculation of seed, plant population and fertilizer requirement	1
9	Study of yield contributing characters and yield estimation of crops	1
10	Identification of weeds in different crops	1
11	Seed germination and viability test of seed	1
12	Practice on time and method of application of manures and fertilizers	1
13	Measurement of soil moisture by gravimetric and volumetric method and bulk density	1
14	Determination of field capacity	1
15	Determination of gross and net irrigation requirement	1
16	Determination of infiltration rate	1
Total		16

References:

1. Rakesh, S. 2022. Textbook on Fundamentals of Agronomy. Shashwat Publication, Bilaspur, Chhattisgarh – 495001
2. Rao V S. 1992. Principles of Weed Science. Oxford and IBH Publishing Co. Ltd. New Delhi.
3. Reddy Yellamanda T and Shankar Reddy G H. 1995. Principles of Agronomy. Kalyani Publishers, Ludhiana.
4. Reddy, S. R. 2008. Principle of Crop Production, Kalyani Publisher, Ludhiana.
5. William L Donn. 1965. Meteorology. McGraw-Hill Book Co. New York.
6. Yawalkar K S and Agarwal J P. 1977. Manures and Fertilizers. Agricultural Horticultural Publishing House, Nagpur.

AGRON 112 FARMING BASED LIVELIHOOD SYSTEMS

3 (2+1)

Objective

1. To make the students aware about farming-based livelihood systems in agriculture
2. To disseminate the knowledge and skill how farming-based systems can be a source of livelihood

Theory

Status of agriculture in India and different states, Income of farmers and rural people in India, Livelihood-Definition, concept and livelihood pattern in urban and rural areas, Different indicators to study livelihood systems. Agricultural livelihood systems (ALS): Meaning, approach, approaches and framework, Definition of farming systems and farming based livelihood systems Prevalent Farming systems in India contributing to livelihood. Types of traditional and modern farming systems. Components of farming system/ farming-based livelihood systems- Crops and cropping systems, Livestock (Dairy, Piggery, Goatry, Poultry, Duckry etc.), Horticultural crops, Agro--forestry systems, Aqua culture Duck/Poultry cum Fish, Dairy cum Fish, Piggery cum Fish etc., small, medium and large enterprises including value chains and secondary enterprises as livelihood components for farmers, Factors affecting integration of various enterprises of farming for livelihood. Feasibility of different farming systems for different agro-climatic zones, Commercial farming-based livelihood models by NABARD, ICAR and other organizations across the country, Case studies on different livelihood enterprises associated with the farming. Risk and success factors in farming-based livelihood systems, Schemes and programs by Central and State Government, Public and Private organizations involved in promotion of farming-based livelihood opportunities. Role of farming-based livelihood enterprises in 21st Century in view of circular economy, green economy, climate change, digitalization and changing life style.

Lecture schedule: Theory

S.N .	Topic	No. of lectures
1.	Status of agriculture in India and different states	1
2.	Income of farmers and rural people in India	1
3.	Livelihood-Definition, concept and livelihood pattern in urban and rural areas, Different indicators to study livelihood systems	2
4.	Agricultural livelihood systems (ALS): Meaning, approach, approaches and framework, Definition of farming systems and farming based livelihood systems	2
5.	Prevalent Farming systems in India contributing to livelihood	1
6.	Types of traditional and modern farming systems	1
7.	Components of farming system/ farming-based livelihood systems- Crops and cropping systems	2
8.	Livestock -Dairy, Piggery,	2
9.	Goatry, Poultry, Duckry etc.	2

10.	Horticultural crops, Agro--forestry systems,	2
11.	Aqua culture Duck/Poultry cum Fish,	2
12.	Dairy cum Fish, Piggery cum Fish etc.	1
13.	Small, medium and large enterprises including value chains and secondary enterprises as livelihood components for farmers	3
14.	Factors affecting integration of various enterprises of farming for livelihood	1
15.	Feasibility of different farming systems for different agro-climatic zones	1
16.	Commercial farming-based livelihood models by NABARD, ICAR and other organizations across the country	2
17.	Case studies on different livelihood enterprises associated with the farming.	1
18.	Risk and success factors in farming-based livelihood systems	1
19.	Schemes and programs by Central and State Government, Public and Private organizations involved in promotion of farming-based livelihood opportunities	2
20.	Role of farming-based livelihood enterprises in 21 st Century in view of circular economy, green economy, climate change, digitalization and changing life style	2
Total		32

Practical

Survey of farming systems and agricultural-based livelihood enterprises, Study of components of important farming- based livelihood models/ systems in different agro-climatic zones, Study of production and profitability of crop based, livestock based, processing based and integrated farmingbased livelihood models, Field visit of innovative farming system models. Visit of agri-based enterprises and their functional aspects for integration of production, processing and distribution sectors and Study of agri-enterprises involved in industry and service sectors (Value Chain Models), Learning about concept of project formulation on farming-based livelihood systems along with cost and profit analysis, Case study of Start-Ups in agri-sectors.

Lecturer schedule: Practical

S. No	Topic	No. of lectures
1	Survey of farming systems and agricultural-based livelihood	1

	enterprises	
2	Study of components of important farming- based livelihood models/systems in different agro-climatic zones	2
3	Study of production and profitability of crop based livelihood models	1
4	Study of production and profitability of livestock based livelihood models	1
5	Study of production and profitability of processing based livelihood models	1
6	Study of production and profitability of integrated farming based livelihood models	1
7	Field visit of innovative farming system models	1
8	Visit of agri-based enterprises and their functional aspects for integration of production, processing and distribution sectors	2
9	Study of agri-enterprises involved in industry and service sectors (Value Chain Models)	2
10	Learning about concept of project formulation on farming-based livelihood systems along with cost and profit analysis	2
11	Case study of Start-Ups in agri-sectors	2
Total		16

Suggested Readings

1. Dixon, J. and A. Gulliver with D. Gibbon. (2001). Farming Systems and Poverty: Improving Farmers' Livelihoods in a Changing World. FAO and World Bank, Rome, Italy and Washington, DC, USA
2. Ashley, C.; Carney, D. (1999). Sustainable Livelihoods: Lessons from Early Experience; Department for International Development: London, UK, Volume 7. [Google Scholar]
3. Reddy, S.R. 2016. Farming System and Sustainable Agriculture, Kalyani Publishers, New Delhi.
4. Panwar et al. 2020. Integrated Farming System models for Agricultural Diversification, Enhanced Income and employment, Indian Council of Agricultural Research, New Delhi.
5. Singh, J.P., et al. 2015. Region Specific Integrated Farming System Models, ICAR-Indian Institute of Farming Systems Research, Modipuram.
6. Walia, S. S. and U. S. Walia, 2020. Farming System and Sustainable Agriculture, Scientific Publishers, Jodhpur, Rajasthan.
7. Livelihood Improvement of Underprivileged Farming Community: Some Experiences from Vaishali, Samastipur, Darbhanga and Munger Districts of Bihar by B. P. Bhatt,

Abhay Kumar, P.K. Thakur, Amitava Dey Ujjwal Kumar, Sanjeev Kumar, B.K Jha, Lokendra Kumar, K. N. Pathak, A. Hassan, S. K. Singh, K. K. Singh and K. M. Singh
ICAR Research Complex for Eastern Region ICAR Parisar, P.O. Bihar Veterinary College, Patna 800 014, Bihar.

HORT-111 FUNDAMENTALS OF HORTICULTURE

3 (2+1)

Objectives

1. To provide knowledge on different branches of horticulture viz. Pomology, Olericulture, Floriculture and landscaping, spices and medicinal plants.
2. To provide knowledge on orchard management, propagation methods, cultural operations and nutrient management of horticultural crops.
3. To provide knowledge on different physiological aspects of horticultural crops.

Theory

Horticulture: Its different branches, importance and scope, Horticulture and botanical classification, Soil and climate for horticultural crops, Nursery management, Plant propagation: methods and propagation structures, Seed dormancy and seed germination, Merits and demerits of sexual and asexual propagation, Stock-scion relationship.

Principles of orchard establishment, Principles and methods of training and pruning of fruit crops, Juvenility and flower bud differentiation, Unfruitfulness in horticultural crops, Pollination, pollinizers and pollinators, Fertilization and parthenocarpy, Importance of bio regulators in horticultural crops, Irrigation and its methods, Fertilizer application in horticultural crops.

Lecture schedule: Theory

S. No.	Topics	No. of lectures
1.	Horticulture: Its different branches, importance and scope	2
2.	Horticulture and botanical classification	2
3.	Soil and climate for horticultural crops	2
4.	Nursery management	2
5.	Plant propagation –Sexual propagation methods	1
6.	Plant propagation – Asexual propagation methods (Cutting, budding, grafting, layering etc.)	3
7.	Merits and demerits of sexual and asexual propagation	1
8.	Stock-scion relationship	1

9.	Propagation structures	2
10.	Seed dormancy and seed germination	2
11.	Principles of orchard establishment	2
12.	Principles and methods of training of fruit crops	2
13.	Principles and methods of pruning of fruit crops	1
14.	Juvenility and flower bud differentiation	1
15.	Unfruitfulness in horticultural crops	1
16.	Pollination, pollinizers and pollinators	2
17.	Fertilization and parthenocarpy	1
18.	Importance of bio regulators in horticultural crops	2
19.	Irrigation and its methods	1
20.	Fertilizer application in horticultural crops	1
Total		32

Practical

Identification and nomenclature of fruit, Layout of an orchard, Pit making and system of planting, Nursery raising techniques of fruit crops, Understanding of plant propagation structures, Propagation through seeds and plant parts, Propagation techniques for horticultural crops, Container, potting mixture, potting and re-potting, Training and pruning methods on fruit crops, Preparation of fertilizer mixture and application, Preparation and application of PGR, Layout of different irrigation systems, Maturity studies, harvesting, grading, packaging and storage.

Lecture schedule: Practical

S. No.	Topics	No. of lectures
1.	Identification and nomenclature of fruit	1
2.	Layout of an orchard	2
3.	Pit making and system of planting	1
4.	Nursery raising techniques of fruit crops	1
5.	Understanding of plant propagation structures	1
6.	Propagation through seeds and plant parts	1
7.	Propagation techniques for horticultural crops	2
8.	Container, potting mixture, potting and re-potting	1

9.	Training and pruning methods on fruit crops	2
10.	Preparation of fertilizer mixture and application	1
11.	Preparation and application of PGR	1
12.	Layout of different irrigation systems	1
13.	Maturity studies, harvesting, grading, packaging and storage	1
Total		16

Suggested readings

1. Chadha K.L., 2024 (22th edition). Handbook of Horticulture, Indian Council of Agriculture Research (ICAR), New-Delhi.
2. Kumar N., 2021 (9th edition). Introduction to Horticulture, MedTech Science Press Scientific Pvt. Ltd.
3. Singh J., 2024 (6th edition). Basic Horticulture, Kalyani Publishers.
4. Yadav, P. K. 2007. Fruit Production Technology, Published by International Book Distributing Company, Lucknow ISBN 81-8189-206-2
5. Yadav, P. K. and Singh, J. 2000. Phalotpadan and Prasanskaran Published in 2000 from Agrobios (India), Jodhpur., ISBN 81-7754-086-6

SSAC-111 FUNDAMENTALS OF SOIL SCIENCE

3 (2+1)

Objective

To impart knowledge on soil genesis, basic soil properties with respect to plant growth

Theory

Soil: Pedological and edaphological concepts. Rocks and minerals, weathering, Silicate clays: constitution and properties, sources of charge, ion exchange, cation and anion exchange capacity and base saturation (after buffering capacity), Soil formation, Soil organic matter, Pedogenic processes, Soil colloids: inorganic and organic, Properties of soil colloids and Ion exchange in soils, Soil profile, soil texture, soil structure. Bulk density and particle density, soil consistency, soil plasticity, soil temperature, soil air, soil water. Soil reaction and buffering capacity. Soil taxonomy, keys to soil orders. Soils of India.

Lecture Schedule: Theory

S. No.	Topics	No. of lectures
1.	Soil: Pedological and edaphological concepts	1
2.	Rocks and minerals, weathering	3

3.	Silicate clays: constitution and properties, sources of charge	3
4.	ion exchange, cation and anion exchange capacity and base saturation (after buffering capacity)	2
5.	Soil formation	2
6.	Soil organic matter	2
7.	Pedogenic processes	2
8.	Soil colloids: inorganic and organic	2
9.	Properties of soil colloids and Ion exchange in soils,	2
10.	Soil profile	1
11.	Soil texture, soil structure	2
12.	Bulk density and particle density, soil consistency, Soil plasticity	1
13.	Soil temperature	2
14.	Soil air	2
15.	Soil water	2
16.	Soil reaction and buffering capacity	1
17.	Soil taxonomy, keys to soil orders. Soils of India.	2
Total theory classes		32

Practical

Study of general properties of minerals, study of minerals-silicate and non-silicate minerals, study of rocks-igneous, sedimentary and metamorphic rocks; study of a soil profile, collection and processing of soil for analysis, study of soil texture-feel method, mechanical analysis, determination particle density, bulk density and soil porosity, determination of soil colour, study of soil structure and aggregate analysis, determination of soil moisture, determination of soil moisture constants field capacity; water holding capacity. Study of infiltration rate of soil, determination of pH and Electrical conductivity of soil.

Lecture Schedule: Practical

S.No.	Topic	No. of lectures
1	Study of general properties of minerals	1
2	Study of minerals-silicate and non-silicate minerals,	1

3	Study of rocks-igneous, sedimentary and metamorphic rocks	1
4	Study of a soil profile	1
5	Collection and processing of soil for analysis	1
6	Study of soil texture-feel method, mechanical analysis	2
7	Determination particle density, bulk density and soil porosity	1
8	Determination of soil colour	1
9	Study of soil structure and aggregate analysis,	2
10	Determination of soil moisture	1
11	Determination of soil moisture constants field capacity; water holding capacity	1
12	Study of infiltration rate of soil	2
13	Determination of pH and Electrical conductivity of soil.	1
	Total practical classes	16

Suggested readings

1. Introductory Soil Science – By Dilip Kumar Das, Kalyani Publishers
2. Soil Fertility and Nutrient Management – By S. S. Singh, Kalyani Publishers
3. Soil Fertility and Fertilizers – By Samuel L. Tisdale, Werner L. Nelson and James D. Beaton, Macmillan Publishing Company, New York
4. The Nature and Properties of Soils – By Harry O. Buckman and Nyle C.

MATH-111

ELEMENTARY MATHEMATICS

1(1+0)

Theory

Algebra: Progressions- Arithmetic, Geometric and Harmonic Progressions. Matrices: Definition of Matrices, Addition, Subtraction, Multiplication, Transpose and Inverse up to 3rd order by adjoint method, Properties of determinants up to 3rd order and their evaluation.

Differential Calculus: Definition, Derivatives of sum, difference, product and quotient of two functions, Methods, Increasing and Decreasing Functions. Application of Differentiation- Growth rate, Average Cost, and Marginal cost, Marginal Cost, Marginal Revenue. Partial differentiation: Homogeneous function, Euler's theorem, Maxima and Minima of the functions of the form $y = f(x)$ and $y = f(x_1, x_2)$.

Lecture Schedule: Theory

S. N.	Topic	No. of lectures
1	Progressions- Arithmetic, Geometric and Harmonic Progressions	2

2	Matrices: Definition of Matrices, Addition, Subtraction	1
3	Matrices: Multiplication, Transpose and Inverse up to 3rd order by adjoint method	2
4	Properties of determinants up to 3rd order and their evaluation.	1
5	Differential Calculus: Definition, Derivatives of sum, difference, product and quotient of two functions	2
6	Increasing and Decreasing Functions	2
7	Application of Differentiation- Growth rate, Average Cost, and Marginal cost, Marginal Cost, Marginal Revenue	2
8	Partial differentiation: Homogeneous function, Euler's theorem	2
9	Maxima and Minima of the functions of the form $y = f(x)$ and $y = f(x_1, x_2)$.	2
Total		16

References

1. Krishi Ganita by Gokhroo and Jain
2. Differential Calculus by Gokhroo.
3. Integral Calculus by Gokhroo.

BIO – 111

INTRODUCTORY BIOLOGY

1(1+0)

Theory

Introduction to the living world, diversity and characteristics of life, origin of life, Evolution and Eugenics. Binomial nomenclature and classification Cell and cell division. Morphology of flowering plants. Seed and seed germination. Plant systematic- viz; Brassicaceae, Fabaceae and Poaceae. Role of animals in agriculture

Lecture Schedule: Theory

S. N.	Topic	No. of lectures
1.	Introduction to the living world, diversity and characteristics of life, origin of life, Evolution and Eugenics.	3
2.	Binomial nomenclature and classification Cell and cell division.	3
3.	Morphology of flowering plants.	2
4.	Seed and seed germination.	3
5.	Plant systematic- viz; Brassicaceae, Fabaceae and Poaceae.	3

6.	Role of animals in agriculture	2
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NCC-I NATIONAL CADET CORPS**1(0+1)**

1. Aims, objectives, organization of NCC and NCC song. DG's cardinal of discipline.
2. Drill-aim, general words of comm. and, attention, stands at ease, stand easy and turning.
3. Sizing, numbering, forming in three ranks, open and close order march and dressing.
4. Saluting at the halt, getting on parade, dismissing and falling out.
5. Marching, length of pace, and time of marching in quick/slow time and halt. Side pace, pace forward and to the rear.
6. Turning on the march and wheeling. Saluting on the march.
7. Marking time, forward march and halt.
8. Changing step, formation of squad and squad drill.
9. Command and control, organization, badges of rank, honours and awards
10. Nation Building-cultural heritage, religions, traditions and customs of India. National integration.

NSS-I NATIONAL SERVICE SCHEME**1(0+1)**

1. Introduction and basic components of NSS: Orientation: history, objectives, principles, symbol, badge; regular programmes under NSS, organizational structure of NSS, code of conduct for NSS volunteers, points to be considered by NSS volunteers awareness about health NSS programmes and activities
2. Concept of regular activities, special camping, day camps, basis of adoption of village/slums, conducting survey, analysing guiding financial patterns of scheme, youth programme/ schemes of GOI, coordination with different agencies and maintenance of diary
3. Understanding youth Definition, profile, categories, issues and challenges of youth; and opportunities for youth who is agent of the social change
4. Community mobilization Mapping of community stakeholders, designing the message as per problems and their culture; identifying methods of mobilization involving youth-adult partnership
5. Social harmony and national integration
6. Indian history and culture, role of youth in nation building, conflict resolution and peace- building
7. Volunteerism and shramdan Indian tradition of volunteerism, its need, importance, motivation and constraints; shramdan as part of volunteerism
8. Citizenship, constitution and human rights
9. Basic features of constitution of India, fundamental rights and duties, human rights, consumer

10. Family and society Concept of family, community (PRIs and other community based organizations) and society

PHED- I PHYSICAL EDUCATION AND YOGA PRACTICES

1(0+1)

1. Teaching of skills of Football demonstration, practice of the skills, correction, involvement in game situation (For girls teaching of Tennikoit)
2. Teaching of different skills of Football demonstration, practice of the skills, correction, involvement in game situation (For girls teaching of Tennikoit)
3. Teaching of advance skills of Football involvement of all the skills in game situation with teaching of rules of the game
4. Teaching of skills of Basketball–demonstration, practice of the skills, correction of skills, involvement in game situation
5. Teaching of skills of Basketball–demonstration, practice of the skills, involvement in game situation
6. Teaching of skills of Basketball – involvement of all the skills in game situation with teaching of rule of the game
7. Teaching of skills of Kabaddi–demonstration, practice of the skills, correction of skills, involvement in game situation
8. Teaching of skills of Kabaddi–demonstration, practice of the skills, correction of skills, involvement in game situation
9. Teaching of advance skills of Kabaddi – involvement of all the skills in game situation with teaching of rule of the game
10. Teaching of skillsof Ball Badminton–demonstration, practice of the skills, correction of skills, involvement in game situation.